

Jeann Victor Batista

Lógica Digital → Lista 1º (Parte A)

a) $(121)_{10} \rightarrow ()_2$

$$\begin{array}{r} 121 \div 2 = 60 \text{ resto } 1 \\ 60 \div 2 = 30 \text{ resto } 0 \\ 30 \div 2 = 15 \text{ resto } 0 \\ 15 \div 2 = 7 \text{ resto } 1 \\ 7 \div 2 = 3 \text{ resto } 1 \\ 3 \div 2 = 1 \text{ resto } 1 \\ 1 \div 2 = 0 \text{ resto } 1 \end{array}$$

$(121)_{10} = (1111001)_2$

b) $(121)_3 = ()_8$

$121 = 1 \times 3^2 + 2 \times 3^1 + 1 \times 3^0 \rightarrow 9 + 6 + 1 = (16)_{10}$

$(16)_{10} \rightarrow (20)_8$

$$\begin{array}{r} 16 \div 8 = 2 \text{ resto } 0 \\ 2 \div 8 = 0 \text{ resto } 2 \end{array}$$

c) $(071)_8 = ()_{16}$

$(071)_8 \rightarrow 7 \times 8^1 + 1 \times 8^0 \rightarrow (57)_{10}$

$(57)_{10} \rightarrow (39)_{16}$

$$\begin{array}{r} 57 \div 16 = 3 \text{ resto } 9 \\ 3 \div 16 = 0 \text{ resto } 3 \end{array}$$

Logo

$(071)_8 \rightarrow (57)_{10} \rightarrow (39)_{16}$

d) $(034)_{16} \rightarrow ()_3$

$(34)_{16} \rightarrow 3 \times 16^1 + 4 \times 16^0 \rightarrow 48 + 4 = 52$

$(52)_{10} \rightarrow (122)_3$

$$\begin{array}{r} 52 \div 3 = 17 \text{ resto } 1 \\ 17 \div 3 = 5 \text{ resto } 2 \\ 5 \div 3 = 1 \text{ resto } 2 \\ 1 \div 3 = 0 \text{ resto } 1 \end{array}$$

Logo

$(034)_{16} \rightarrow (52)_{10} \rightarrow (122)_3$

e) $(AF)_{16} = ()_{10}$

$10 \times 16^1 + 15 \times 16^0 \rightarrow 160 + 15 = 175$

Logo

$(AF)_{16} \rightarrow (175)_{10}$

f) $(0009A)_{16} = ()_8$

$9 \times 16^1 + 10 \times 16^0 = (154)_{10}$

$(154)_{10} \rightarrow (232)_8$

$$\begin{array}{r} 154 \div 8 = 19 \text{ resto } 2 \\ 19 \div 8 = 2 \text{ resto } 3 \\ 2 \div 8 = 0 \text{ resto } 2 \end{array}$$

Logo

$(0009A)_{16} = (154)_{10} \rightarrow (232)_8$

g) $(111011)_4 = ()_8$

$1 \times 4^5 + 1 \times 4^4 + 1 \times 4^3 + 0 \times 4^2 + 1 \times 4^1 + 1 \times 4^0$

$1024 + 256 + 64 + 0 + 4 + 1 = (1349)_{10}$

$(1349)_{10}$

$(1349)_{10} \rightarrow (2505)_8$

$$\begin{array}{r} 1349 \div 8 = 168 \text{ resto } 5 \\ 168 \div 8 = 21 \text{ resto } 0 \\ 21 \div 8 = 2 \text{ resto } 5 \\ 2 \div 8 = 0 \text{ resto } 2 \end{array}$$

Logo

$(111011)_4 = (1349)_{10} \rightarrow (2505)_8$

$$b) (00009A)_{16} = ()_2$$

$$9 \times 16^1 + 10 \times 16^0 = (154)_{10}$$

$$\begin{array}{r} 154 \div 2 = 77 \text{ resto } 0 \\ 77 \div 2 = 38 \text{ resto } 1 \\ 38 \div 2 = 19 \text{ resto } 0 \\ 19 \div 2 = 9 \text{ resto } 1 \\ 9 \div 2 = 4 \text{ resto } 1 \\ 4 \div 2 = 2 \text{ resto } 0 \\ 2 \div 2 = 1 \text{ resto } 0 \\ 1 \div 2 = 0 \text{ resto } 1 \end{array}$$

$$(9A)_{16} = (154)_{10} \rightarrow (110011010)_2$$

$$2) a) +(121)_{10} = ()_2$$

$$\begin{array}{r} 121 \div 2 = 60 \text{ resto } 1 \\ 60 \div 2 = 30 \text{ resto } 0 \\ 30 \div 2 = 15 \text{ resto } 0 \\ 15 \div 2 = 7 \text{ resto } 1 \\ 7 \div 2 = 3 \text{ resto } 1 \\ 3 \div 2 = 1 \text{ resto } 1 \\ 1 \div 2 = 0 \text{ resto } 1 \end{array}$$

$$(1111001)_2$$

$$(1000000001111001)_2 \text{ em 16 bit}$$

$$b) -(121)_3 = ()_2$$

$$1 \times 3^2 + 2 \times 3^1 + 1 \times 3^0 = 9 + 6 + 1 = (16)_{10}$$

$$\begin{array}{r} 16 \div 2 = 8 \text{ resto } 0 \\ 8 \div 2 = 4 \text{ resto } 0 \\ 4 \div 2 = 2 \text{ resto } 0 \\ 2 \div 2 = 1 \text{ resto } 0 \\ 1 \div 2 = 0 \text{ resto } 1 \end{array}$$

$$(100001)_2 \rightarrow (1000000000010000)_2 \text{ em 16 bit}$$

$$c) -(071)_8 = ()_2$$

$$(071)_8 = 7 \times 8^1 + 1 \times 8^0 = 57$$

$$\begin{array}{r} 57 \div 2 = 28 \text{ resto } 1 \\ 28 \div 2 = 14 \text{ resto } 0 \\ 14 \div 2 = 7 \text{ resto } 0 \\ 7 \div 2 = 3 \text{ resto } 1 \\ 3 \div 2 = 1 \text{ resto } 1 \\ 1 \div 2 = 0 \text{ resto } 1 \end{array}$$

$$(111001)_2 \rightarrow (100000000111001)_2 \text{ em 16 bit}$$

$$d) (034)_{16} \rightarrow ()_2$$

$$(034)_{16} = 3 \times 16^1 + 4 \times 16^0 = 48 + 4 = (52)_{10}$$

$$\begin{array}{r} 52 \div 2 = 26 \text{ resto } 0 \\ 26 \div 2 = 13 \text{ resto } 0 \\ 13 \div 2 = 6 \text{ resto } 1 \\ 6 \div 2 = 3 \text{ resto } 0 \\ 3 \div 2 = 1 \text{ resto } 1 \\ 1 \div 2 = 0 \text{ resto } 1 \end{array}$$

$$(110100)_2 \rightarrow (100000000110100)_2 \text{ em 16 bit}$$

$$e) -(034)_8 = ()_2$$

$$(034)_8 = 3 \times 8^1 + 4 \times 8^0 = 28$$

$$\begin{array}{r} 28 \div 2 = 14 \text{ resto } 0 \\ 14 \div 2 = 7 \text{ resto } 0 \\ 7 \div 2 = 3 \text{ resto } 1 \\ 3 \div 2 = 1 \text{ resto } 1 \\ 1 \div 2 = 0 \text{ resto } 1 \end{array}$$

$$(034)_8 = (11100)_2 \rightarrow (10000000011100)_2 \text{ em 16 bit}$$

$$f) -(AF)_{16} \rightarrow ()_2$$

$$10 \times 16^1 + 15 \times 16^0 = (175)_{10}$$

$$\begin{array}{r} 175 \div 2 = 87 \text{ resto } 1 \\ 87 \div 2 = 43 \text{ resto } 1 \\ 43 \div 2 = 21 \text{ resto } 1 \\ 21 \div 2 = 10 \text{ resto } 1 \\ 10 \div 2 = 5 \text{ resto } 0 \\ 5 \div 2 = 2 \text{ resto } 1 \\ 2 \div 2 = 1 \text{ resto } 0 \\ 1 \div 2 = 0 \text{ resto } 1 \end{array}$$

$$-(AF)_{16} = (10101111)_2 \rightarrow (1000000010101111)_2 \text{ em 16 bit}$$

$$-(0009A)_{16} = ()_2 \quad \left\{ \begin{array}{l} 154 \div 2 \\ 77 \div 2 \\ 38 \div 2 \\ 19 \div 2 \\ 9 \div 2 \\ 4 \div 2 \\ 2 \div 2 \\ 1 \div 2 \end{array} \right.$$

$$9 \times 16^1 + 10 \times 16^0 = (154)_{10}$$

$$(009A)_{16} = (10011010)_2 \rightarrow 11000000010011010 \text{ em 16 bit}$$

$$H) + (111011)_2 \rightarrow ()_2$$

$$1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = (1349)_{10}$$

$$11010100010112$$

$$10000001010100010112 \text{ em 16 bit}$$

$$\textcircled{1} + (00009A)_{16} = ()_2$$

$$9 \times 16^1 + 10 \times 16^0 = (154)_{10}$$

$$(10111010)_{\text{base } 2} \rightarrow 10000000010111010 \text{ em 16 bit}$$

$$\textcircled{3} a) (12)_{10} \rightarrow ()_2$$

$$12 \div 2 = 6 \text{ resto } 0$$

$$6 \div 2 = 3 \text{ resto } 0$$

$$3 \div 2 = 1 \text{ resto } 1$$

$$1 \div 2 = 0 \text{ resto } 1$$

$$(1100)_2$$

$$1111001 \text{ Byte}$$

$$b) -(12)_{10} \rightarrow ()_2$$

$$12 = 1 \times 2^3 + 2 \times 2^1 + 1 \times 2^0 = (1100)_{10}$$

$$000000000010000$$

$$11111111110111$$

$$11111111110000 + 1$$

$$11111111110001$$

$$\textcircled{C} - (071)_8 \rightarrow ()_2$$

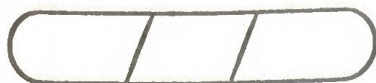
$$071 \rightarrow 7 \times 8^1 + 1 \times 8^0 \rightarrow 57$$

$$111001 \rightarrow 0000000000111001$$

$$1111111111000110$$

$$+1$$

$$1111111111000111$$



d) $(034)_{16} \rightarrow 1 \cdot 12 \rightarrow 52$
 $3 \times 16^1 + 4 \times 16^0 \rightarrow 52$

Long division steps:
 $52 \div 2 = 26$
 $26 \div 2 = 13$
 $13 \div 2 = 6 \text{ R } 1$
 $6 \div 2 = 3$
 $3 \div 2 = 1 \text{ R } 1$
 $1 \div 2 = 0 \text{ R } 1$

$110100 \rightarrow 0000000000110100$

e) $(034)_8 = 1 \cdot 12$
 $(34) \rightarrow 3 \times 8^1 + 4 \times 8^0 \rightarrow 28$

Long division steps:
 $28 \div 2 = 14$
 $14 \div 2 = 7$
 $7 \div 2 = 3 \text{ R } 1$
 $3 \div 2 = 1 \text{ R } 1$
 $1 \div 2 = 0 \text{ R } 1$

$11100 \rightarrow 000000000011100$
 $000000000011100 + 1 = 000000000011101$

f) $(-AF)_{16}$
 $10 \times 16^1 + 15 \times 16^0 \rightarrow 175$

Long division steps:
 $175 \div 2 = 87 \text{ R } 1$
 $87 \div 2 = 43 \text{ R } 1$
 $43 \div 2 = 21 \text{ R } 1$
 $21 \div 2 = 10 \text{ R } 1$
 $10 \div 2 = 5 \text{ R } 0$
 $5 \div 2 = 2 \text{ R } 1$
 $2 \div 2 = 1 \text{ R } 0$
 $1 \div 2 = 0 \text{ R } 1$

$10101111 \rightarrow (000000010101111) \rightarrow (111111101010000) + 1 = 11111110100001$

g) $(0009A)_{16} = 1 \cdot 12$
 $9 \times 16^1 + 10 \times 16^0 \rightarrow 154$

Long division steps:
 $154 \div 2 = 77$
 $77 \div 2 = 38 \text{ R } 1$
 $38 \div 2 = 19$
 $19 \div 2 = 9 \text{ R } 1$
 $9 \div 2 = 4 \text{ R } 1$
 $4 \div 2 = 2$
 $2 \div 2 = 1$
 $1 \div 2 = 0 \text{ R } 1$

$(10011010)_2 \rightarrow (000000010011010) \rightarrow (11111110010101) + 1 = 11111110010110$

h) $(1111011)_2$
 $1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 32 + 16 + 8 + 0 + 2 + 1 = 59$

Long division steps:
 $59 \div 2 = 29 \text{ R } 1$
 $29 \div 2 = 14 \text{ R } 1$
 $14 \div 2 = 7$
 $7 \div 2 = 3 \text{ R } 1$
 $3 \div 2 = 1 \text{ R } 1$
 $1 \div 2 = 0 \text{ R } 1$

$110101000101 \rightarrow 10000000101000101$

$(100009A)_{16} = ()_2 \rightarrow 15412$
 $9 \times 16^1 + 10 \times 16^0 = 154$
 $10011010 \rightarrow 0000000010011010$

a) $11100110_2 \rightarrow 00011010_2 \rightarrow 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$
 $16 + 8 + 0 + 2 + 0 = 26 \rightarrow -26$

b) $101101101_2 \rightarrow 0 \times 2^7 + 1 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$
 $0 + 64 + 32 + 0 + 8 + 4 + 0 + 1 = 109$

c) $(01010101)_2 \rightarrow ()_{10} \rightarrow 0 \times 2^7 + 1 \times 2^6 + 0 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$
 $0 + 64 + 0 + 16 + 0 + 4 + 0 + 1 = 85$

d) $(01111110)_2 \rightarrow ()_{10} \rightarrow 0 \times 2^7 + 1 \times 2^6 + 1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$
 $0 + 64 + 32 + 16 + 8 + 4 + 2 + 0 = 126$

e) $(10000001)_2 \rightarrow ()_{10}$
 $(01111110)_2 \rightarrow (01111111) \rightarrow 0 \times 2^7 + 1 \times 2^6 + 1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$
 $0 + 64 + 32 + 16 + 8 + 4 + 2 + 1 \rightarrow -127$

f) $(10000000)_2 \rightarrow ()_{10}$
 $01111111 \rightarrow 10000000 \rightarrow 1 \times 2^7 \rightarrow -128$

g) $(10100001)_2 \rightarrow ()_{10}$
 $01011110 \rightarrow 0 \times 2^7 + 1 \times 2^6 + 0 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$
 $0 \quad 64 \quad 0 \quad 16 \quad 8 \quad 4 \quad 2 \quad 1 = -95$

$$h) 1000000011_2 \rightarrow 110 \rightarrow 1 \times 2^0 \rightarrow (1)_1$$

$$i) (11111110)_2 \rightarrow$$

$$\begin{array}{r} 00000001 \\ +1 \\ \hline 10 \end{array}$$

5)

$$a) (11100110)_2 \rightarrow 128 + 64 + 32 + 4 + 2 = 230 \rightarrow (102)_{10}$$

$$\begin{array}{r} 230 \\ -128 \\ \hline 102 \end{array}$$

$$b) (01101101)_2 \rightarrow 0 + 64 + 32 + 8 + 4 + 1 = 109 \rightarrow 109 \rightarrow (-19)_{10}$$

$$\begin{array}{r} 109 \\ -128 \\ \hline \end{array}$$

$$c) (01010101)_2 \rightarrow 64 + 16 + 4 + 1 = 85 \rightarrow 85 \rightarrow (-43)_{10}$$

$$\begin{array}{r} -128 \\ +85 \\ \hline -43 \end{array}$$

$$d) 01111110 \rightarrow 64 + 32 + 16 + 8 + 4 + 2 = 126 \rightarrow (-2)_{10}$$

$$\begin{array}{r} 126 \\ -128 \\ \hline \end{array}$$

$$e) 10000001 \rightarrow 128 + 1 \rightarrow 129 \rightarrow 1$$

$$\begin{array}{r} 129 \\ -128 \\ \hline 1 \end{array}$$

$$f) 10000000 \rightarrow 128 \rightarrow 128 \rightarrow 0$$

$$\begin{array}{r} 128 \\ -128 \\ \hline 0 \end{array}$$

$$g) (110100001)_2 \rightarrow 128 + 32 + 1 = 161 \rightarrow (33)_{10}$$

$$\begin{array}{r} 161 \\ -128 \\ \hline 33 \end{array}$$

$$h) 00000001 \rightarrow 1 \rightarrow 1 \rightarrow -127$$

$$\begin{array}{r} 1 \\ -128 \\ \hline -127 \end{array}$$

$$i) (11111110)_2 \rightarrow 128 + 64 + 32 + 16 + 8 + 4 + 2 = 254 \rightarrow (-2)_{10}$$

$$\begin{array}{r} 254 \\ -128 \\ \hline 126 \end{array}$$

$$\begin{array}{r} 11100110 \\ + 00001101 \\ \hline 11101111 \end{array}$$

b

$$\begin{array}{r} 14 \overline{) 2} \\ 0 \ 7 \ 2 \\ \hline 1 \ 3 \end{array} \rightarrow \begin{array}{r} 3 \overline{) 2} \\ 1 \ 1 \ 2 \\ \hline 1 \ 0 \end{array} \rightarrow \begin{array}{r} 103 \overline{) 2} \\ 1 \ 5 \ 1 \ 2 \\ \hline 1 \ 2 \ 5 \ 2 \\ \hline 1 \ 12 \end{array} \rightarrow \begin{array}{r} 12 \overline{) 2} \\ 0 \ 6 \ 2 \\ \hline 0 \ 3 \ 2 \\ \hline 1 \ 1 \ 2 \\ \hline 1 \ 0 \end{array} \rightarrow \begin{array}{r} 01100111 \rightarrow 10011000 \\ + 1 \\ \hline 10011001 \end{array}$$

$$\begin{array}{r} 10011001 \\ + 00001110 \\ \hline 10100111 \rightarrow -89 \end{array}$$

c

$$\begin{array}{r} 11100110 \rightarrow 128 - 64 - 32 - 4 - 2 = -26 \\ + 11111101 \rightarrow 128 - 64 - 32 - 16 - 8 - 4 - 1 = -3 \\ \hline 11101001 \end{array}$$

$$\begin{array}{r} -26 - (-3) = -26 + 3 = -23 \rightarrow \begin{array}{r} 23 \overline{) 2} \\ 1 \ 1 \ 1 \ 2 \\ \hline 1 \ 5 \end{array} \rightarrow \begin{array}{r} 5 \overline{) 2} \\ 1 \ 2 \ 1 \ 2 \\ \hline 1 \ 1 \ 2 \\ \hline 1 \ 0 \end{array} \rightarrow \begin{array}{r} 00010111 \\ 11101000 \\ \hline 11101001 \end{array}$$

d

$$\begin{array}{r} -45 \\ + 103 \\ \hline \end{array} \rightarrow \begin{array}{r} 148 \overline{) 2} \\ 0 \ 7 \ 4 \ 2 \\ \hline 1 \ 37 \end{array} \rightarrow \begin{array}{r} 37 \overline{) 2} \\ 1 \ 18 \ 2 \\ \hline 1 \ 9 \end{array} \rightarrow \begin{array}{r} 9 \overline{) 2} \\ 1 \ 4 \ 2 \\ \hline 1 \ 12 \end{array} \rightarrow \begin{array}{r} 21 \overline{) 2} \\ 0 \ 1 \ 2 \\ \hline 1 \ 0 \end{array} \rightarrow \begin{array}{r} 010010100 \rightarrow 101101011 \\ + 1 \\ \hline 101101100 \end{array}$$

$$\begin{array}{r} -45 \overline{) 2} \\ 1 \ 27 \ 2 \\ \hline 1 \ 11 \ 2 \\ \hline 1 \ 5 \end{array} \rightarrow \begin{array}{r} 5 \overline{) 2} \\ 1 \ 2 \ 2 \\ \hline 1 \ 1 \ 2 \\ \hline 1 \ 0 \end{array} \rightarrow \begin{array}{r} 00101101 \rightarrow 11010010 \\ + 1 \\ \hline 11010011 \end{array} \rightarrow \begin{array}{r} 11010011 \\ 10011001 \\ \hline 10101010 \end{array}$$

$$\begin{array}{r} 103 \overline{) 2} \\ 1 \ 5 \ 1 \ 2 \\ \hline 1 \ 25 \ 2 \\ \hline 1 \ 13 \ 2 \\ \hline 1 \ 6 \end{array} \rightarrow \begin{array}{r} 6 \overline{) 2} \\ 1 \ 3 \ 2 \\ \hline 1 \ 1 \ 2 \\ \hline 1 \ 0 \end{array} \rightarrow \begin{array}{r} 01100111 \rightarrow 10011000 \\ + 1 \\ \hline 10011001 \end{array}$$

e

$$\begin{array}{r} 11100110 \rightarrow -26 \\ \times 00001101 \rightarrow 13 \\ \hline 11100110 \\ 100000000 \\ 11100110 \\ 11100110 \\ \hline \end{array}$$

$$1110101110 \rightarrow -338$$



F) $11100110 \rightarrow 2187 + 729 + 243 + 93 = 3171$

$$\begin{array}{r} 3171 \\ - 37 \\ \hline 3134 \end{array}$$

$$00001101 \rightarrow 27 + 9 + 1 = 37$$

$$3134 \div 2$$

$$\begin{array}{r} 1567 \div 2 \\ \hline 1 \end{array}$$

$$483 \div 2$$

$$\begin{array}{r} 241 \div 2 \\ \hline 1 \end{array}$$

$$195 \div 2$$

$$\begin{array}{r} 97 \div 2 \\ \hline 1 \end{array}$$

$$48 \div 2$$

$$\begin{array}{r} 24 \div 2 \\ \hline 0 \end{array}$$

$$12 \div 2$$

$$\begin{array}{r} 6 \div 2 \\ \hline 0 \end{array}$$

$$3 \div 2$$

$$\begin{array}{r} 1 \div 2 \\ \hline 0 \end{array}$$

$1110000111110 \rightarrow 3137$

$6 \div 2 \text{ em } 12 \div 2 = 14 \div 2 \rightarrow 7 \div 2 \rightarrow 3 \div 2 \rightarrow 1 \div 2$

$100001110 \rightarrow 11110001$

$$103 \div 2$$

$$\begin{array}{r} 51 \div 2 \\ \hline 1 \end{array}$$

$$25 \div 2$$

$$\begin{array}{r} 12 \div 2 \\ \hline 0 \end{array}$$

$$6 \div 2$$

$$\begin{array}{r} 3 \div 2 \\ \hline 1 \end{array}$$

$$1 \div 2$$

$$\begin{array}{r} 0 \div 2 \\ \hline 0 \end{array}$$

$$01100111$$

$$11110010$$

$$+ 01100111$$

$$\hline 101011001$$

mulit

$$1 \rightarrow 01011001$$

H) $121_{(8)} = 1 \times 8^1 + 2 \times 8^0 = 12 \div 2 \rightarrow 6 \div 2 \rightarrow 3 \div 2 \rightarrow 1 \div 2$

$00001100 \rightarrow 11110011$

$103 \rightarrow 1 \times 8^2 + 0 \times 8^1 + 3 \times 8^0 \rightarrow 64 + 3 \rightarrow 67$

$$67 \div 2$$

$$\begin{array}{r} 33 \div 2 \\ \hline 1 \end{array}$$

$$16 \div 2$$

$$\begin{array}{r} 8 \div 2 \\ \hline 0 \end{array}$$

$$4 \div 2$$

$$\begin{array}{r} 2 \div 2 \\ \hline 0 \end{array}$$

$$1 \div 2$$

$$\begin{array}{r} 0 \div 2 \\ \hline 0 \end{array}$$

$$01000011$$

$$11110100$$

$$+ 01000011$$

$$\hline 10011011$$

$$\rightarrow 00110111 \rightarrow (67)_{10}$$

	SM	C1	C2	Excesso de 2^{n-1} Polarizado
0	10000000 / 00000000	00000000 / 11111111	00000000 / 11111111	0 127,0 (1)
+1	00000001 / 10000001	00000001 / 11111110	00000001 / 11111111	128,0
+2	00000010 / 10000010	00000010 / 11111101	00000010 / 11111110	129,0
+3	00000011 / 10000011	00000011 / 11111100	00000011 / 11111101	130,0
+4	00000100 / 10000100	00000100 / 11111011	00000100 / 11111100	131,0

- 8) a) Overflow significa ultrapassar o limite de representação.
Em sua utilidade prática é ganhar precisão e menor perda de info.
- b)

c) IEEE = Instituto de engenheiros eletricitas e eletronicos
O escopo das atividades que essa entidade engloba é o avanço da tecnologia em benefício da humanidade.

d) A facilidade das operações, e a necessidade menos caracteres para a representação são duas das principais motivos pelos quais é optado.

e) Apenas uma representação de 0, e não no ~~comple~~ excesso de 2^{n-1} existe a vantagem de se dar ou mais números positivos ou negativos.

f) É basicamente o valor absoluto de um número (não conta o sinal).

g) Overflow → Ocorre quando o resultado de uma operação aritmética é muito grande para ser representado dentro do intervalo de valores válidos para o formato de ponto flutuante.

Underflow → Ocorre quando o resultado de uma operação aritmética é muito próximo de zero para ser representado com precisão dentro do intervalo de valores válidos para o formato de ponto flutuante.

h) Por conta das dízimas periódicas, juntamente por conta do tamanho da representação.



127
6

9) a) $(101, 324)_{10} \rightarrow$

$\begin{array}{r} 10112 \\ \textcircled{1} 50 \overline{) 12} \\ \textcircled{0} 25 \end{array}$	$\rightarrow$	$\begin{array}{r} 2512 \\ \textcircled{1} 12 \overline{) 12} \\ \textcircled{0} 6 \end{array}$	$\rightarrow$	$\begin{array}{r} 612 \\ \textcircled{0} 3 \overline{) 12} \\ \textcircled{1} 1 \end{array}$	$\rightarrow$	$\begin{array}{r} 112 \\ \textcircled{1} 0 \end{array}$
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1100101

$\begin{array}{r} 324 \\ \times 2 \\ \hline 648 \end{array}$	$\begin{array}{r} 68 \\ \times 2 \\ \hline 136 \end{array}$	$\begin{array}{r} 36 \\ \times 2 \\ \hline 72 \end{array}$	$\begin{array}{r} 72 \\ \times 2 \\ \hline 144 \end{array}$	$\begin{array}{r} 44 \\ \times 2 \\ \hline 88 \end{array}$	$\begin{array}{r} 88 \\ \times 2 \\ \hline 176 \end{array}$	$\begin{array}{r} 76 \\ \times 2 \\ \hline 152 \end{array}$	$\begin{array}{r} 52 \\ \times 2 \\ \hline 104 \end{array}$	$\begin{array}{r} 24 \\ \times 2 \\ \hline 48 \end{array}$	$\begin{array}{r} 16 \\ \times 2 \\ \hline 32 \end{array}$	$\begin{array}{r} 32 \\ \times 2 \\ \hline 64 \end{array}$
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$\begin{array}{r} 28 \\ \times 2 \\ \hline 56 \end{array}$	$\begin{array}{r} 56 \\ \times 2 \\ \hline 112 \end{array}$	$\begin{array}{r} 12 \\ \times 2 \\ \hline 24 \end{array}$	$\begin{array}{r} 24 \\ \times 2 \\ \hline 48 \end{array}$	$\begin{array}{r} 48 \\ \times 2 \\ \hline 96 \end{array}$	$\begin{array}{r} 96 \\ \times 2 \\ \hline 192 \end{array}$	$\begin{array}{r} 84 \\ \times 2 \\ \hline 168 \end{array}$	$\begin{array}{r} 68 \\ \times 2 \\ \hline 136 \end{array}$
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01010111000010100011...

8 bits

010000110010101010101010101010

b) $\begin{array}{r} 9112 \\ \textcircled{1} 45 \overline{) 12} \\ \textcircled{1} 22 \end{array}$ $\begin{array}{r} 2212 \\ \textcircled{0} 11 \overline{) 12} \\ \textcircled{1} 5 \end{array}$ $\begin{array}{r} 512 \\ \textcircled{0} 1 \overline{) 12} \\ \textcircled{0} 1 \end{array}$ $\begin{array}{r} 112 \\ \textcircled{1} 0 \end{array}$ 01,011011

$\begin{array}{r} 5 \\ \times 2 \\ \hline 10 \end{array}$	$\begin{array}{r} 10 \\ \times 2 \\ \hline 20 \end{array}$	$\begin{array}{r} 20 \\ \times 2 \\ \hline 40 \end{array}$	$\begin{array}{r} 40 \\ \times 2 \\ \hline 80 \end{array}$	$\begin{array}{r} 80 \\ \times 2 \\ \hline 160 \end{array}$	$\begin{array}{r} 60 \\ \times 2 \\ \hline 120 \end{array}$	$\begin{array}{r} 20 \\ \times 2 \\ \hline 40 \end{array}$	$\begin{array}{r} 10 \\ \times 2 \\ \hline 20 \end{array}$	$\begin{array}{r} 30 \\ \times 2 \\ \hline 60 \end{array}$	$\begin{array}{r} 160 \\ \times 2 \\ \hline 320 \end{array}$	$\begin{array}{r} 20 \\ \times 2 \\ \hline 40 \end{array}$	$\begin{array}{r} 40 \\ \times 2 \\ \hline 80 \end{array}$	$\begin{array}{r} 80 \\ \times 2 \\ \hline 160 \end{array}$
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8 bits

11000010101101100001100110011010

c) -34712 $\begin{array}{r} 8612 \\ \textcircled{1} 173 \overline{) 12} \\ \textcircled{1} 96 \end{array}$ $\begin{array}{r} 2112 \\ \textcircled{0} 43 \overline{) 12} \\ \textcircled{1} 21 \end{array}$ $\begin{array}{r} 512 \\ \textcircled{1} 10 \overline{) 12} \\ \textcircled{0} 5 \end{array}$ $\begin{array}{r} 512 \\ \textcircled{1} 2 \overline{) 12} \\ \textcircled{0} 8 \end{array}$ 1,0101011

8 casa $\begin{array}{r} 127 \\ \times 5 \\ \hline 135 \end{array}$

$\begin{array}{r} 92 \\ \times 2 \\ \hline 184 \end{array}$	$\begin{array}{r} 184 \\ \times 2 \\ \hline 368 \end{array}$	$\begin{array}{r} 368 \\ \times 2 \\ \hline 736 \end{array}$	$\begin{array}{r} 736 \\ \times 2 \\ \hline 1472 \end{array}$	$\begin{array}{r} 1472 \\ \times 2 \\ \hline 2944 \end{array}$	$\begin{array}{r} 2944 \\ \times 2 \\ \hline 5888 \end{array}$	$\begin{array}{r} 5888 \\ \times 2 \\ \hline 11776 \end{array}$	$\begin{array}{r} 11776 \\ \times 2 \\ \hline 23552 \end{array}$	$\begin{array}{r} 23552 \\ \times 2 \\ \hline 47104 \end{array}$	$\begin{array}{r} 47104 \\ \times 2 \\ \hline 94208 \end{array}$	$\begin{array}{r} 94208 \\ \times 2 \\ \hline 188416 \end{array}$	$\begin{array}{r} 188416 \\ \times 2 \\ \hline 376832 \end{array}$	$\begin{array}{r} 376832 \\ \times 2 \\ \hline 753664 \end{array}$	$\begin{array}{r} 753664 \\ \times 2 \\ \hline 1507328 \end{array}$
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8 bits

tilibra 11000011101011011000101111010010

A, Vai ao contrario

0 → 0000000 como é entre 0 e -1

0,622	244	488	976	1952	3904	7808	15616	31232	62464	124928	249856	499712
$\times 2$	$\times 2$	$\times 2$	$\times 2$	$\times 2$	$\times 2$	$\times 2$	$\times 2$	$\times 2$	$\times 2$	$\times 2$	$\times 2$	$\times 2$
1,244	0,488	0,976	1,952	3,904	7,808	15,616	31,232	62,464	124,928	249,856	499,712	999,424

424	848	1696	3392	6784	13568	27136	54272	108544	217088	434176	868352	1736704
$\times 2$	$\times 2$	$\times 2$	$\times 2$	$\times 2$	$\times 2$	$\times 2$	$\times 2$	$\times 2$	$\times 2$	$\times 2$	$\times 2$	$\times 2$
848	1696	3392	6784	13568	27136	54272	108544	217088	434176	868352	1736704	3473408

8

1011111100011110011101101100100

e) 1322 → 3312 → 812 → 212

$\begin{matrix} \textcircled{0} & 66 & 2 \\ \textcircled{0} & 33 & \end{matrix}$
→
 $\begin{matrix} \textcircled{1} & 16 & 2 \\ \textcircled{0} & 8 & \end{matrix}$
→
 $\begin{matrix} \textcircled{0} & 4 & 2 \\ \textcircled{0} & 2 & \end{matrix}$
→
 $\begin{matrix} \textcircled{0} & 1 & 2 \\ \textcircled{1} & 0 & \end{matrix}$

10000100

+ 128 →

40	50	80	60	20	40	80	60	20
$\times 2$	$\times 2$	$\times 2$	$\times 2$	$\times 2$	$\times 2$	$\times 2$	$\times 2$	$\times 2$
0,90	1,80	1,60	1,20	0,40	0,80	1,60	1,20	0,40

0 110 ...

008 bits

01000011000001000111001100110011